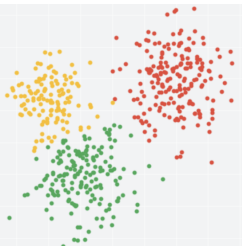


k-Medians Algorithm

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k -Means algorithm

The objective function

$$\min_{\bar{Y}_1, \dots, \bar{Y}_k} \sum_{i=1}^k \sum_{\bar{X} \in C_i} \|\bar{X} - \bar{Y}_i\|^2$$

where C_i consists of the objects that are closest to $\bar{Y}_i$.

We want to minimise the total **squared Euclidean distance** between data objects and their cluster representatives $\bar{Y}_1, \dots, \bar{Y}_k$

This objective function is called the *within cluster sum of squares (WCSS)* objective

k -Means algorithm

What if **Manhattan distance** is more appropriate for our application than the **Euclidean distance**?

Manhattan distance (or L^1 distance)

$$\text{ManDist}(\bar{X}, \bar{Y}) = \|\bar{X} - \bar{Y}\|_1 = \sum_{i=1}^d |x_i - y_i|$$

k -Medians algorithm

The objective function

$$\min_{\bar{Y}_1, \dots, \bar{Y}_k} \sum_{i=1}^k \sum_{\bar{X} \in C_i} \|\bar{X} - \bar{Y}_i\|_1$$

where C_i consists of the objects that are L^1 -closest to $\bar{Y}_i$.

We want to minimise the total L^1 -distance between data objects and their cluster representatives $\bar{Y}_1, \dots, \bar{Y}_k$.

k -Medians algorithm

$$\arg \min_{\bar{Y}} \sum_{\bar{X} \in C} \|\bar{X} - \bar{Y}\|_1$$

$$\sum_{\bar{X} \in C} \|\bar{X} - \bar{Y}\|_1 = \sum_{\bar{X} \in C} \sum_{i=1}^d |\bar{X}^{(i)} - \bar{Y}^{(i)}| = \sum_{i=1}^d \sum_{\bar{X} \in C} |\bar{X}^{(i)} - \bar{Y}^{(i)}|$$

Since the coordinates are independent, to minimise the latter sum we can minimise each of its terms independently. More specifically, if for $i = 1, \dots, d$, the number $\bar{Y}^{(i)}$ minimises

$$\sum_{\bar{X} \in C} |\bar{X}^{(i)} - \bar{Y}^{(i)}|,$$

then

$$\bar{Y}' = (\bar{Y}^{(1)}, \bar{Y}^{(2)}, \dots, \bar{Y}^{(d)})$$

minimises

$$\sum_{\bar{X} \in C} \|\bar{X} - \bar{Y}\|_1.$$

k -Medians algorithm

Given s numbers $\bar{X}_1^{(i)}, \bar{X}_2^{(i)}, \dots, \bar{X}_s^{(i)}$, the number

$$\bar{Y}^{(i)} = \text{median}(\bar{X}_1^{(i)}, \bar{X}_2^{(i)}, \dots, \bar{X}_s^{(i)})$$

minimises

$$\sum_{t=1}^s \left| \bar{X}_t^{(i)} - \bar{Y}^{(i)} \right|.$$

Think about them as the i -th coordinates of the objects in cluster C .

Definition

A median of a sequence of numbers is any value such that at most half of the values is less than the proposed median and at most half is greater than the proposed median.

k -Medians algorithm

Hence an object

$$\bar{Y}' = (\bar{Y}^{(1)}, \bar{Y}^{(2)}, \dots, \bar{Y}^{(d)})$$

that minimises

$$\arg \min_{\bar{Y}} \sum_{\bar{X} \in C} \|\bar{X} - \bar{Y}\|_1$$

is a **median object** of (the objects in) the cluster C ;

where $\bar{Y}^{(i)}$ is the median of the i -th coordinates of the objects in the cluster C .

k -Medians algorithm

k-MediansClustering (Number of clusters: k , Dataset: $\{\bar{X}_1, \dots, \bar{X}_n\}$)

1. Initialisation phase

Choose k cluster representatives $\bar{Y}_1, \dots, \bar{Y}_k$ from the dataset randomly.

2. Assignment phase

Assign all objects in the dataset to the L^1 -closest representative.
The resulting clusters: $C_1, \dots, C_k$.

3. Optimisation phase

Compute the new representatives $\bar{Y}_1, \dots, \bar{Y}_k$ as the **medians** of the current clusters $C_1, \dots, C_k$.

Repeat Phases 2 and 3 until convergence.

(Convergence is either “no objects have moved among clusters” or a fixed number of iterations specified by the user.)